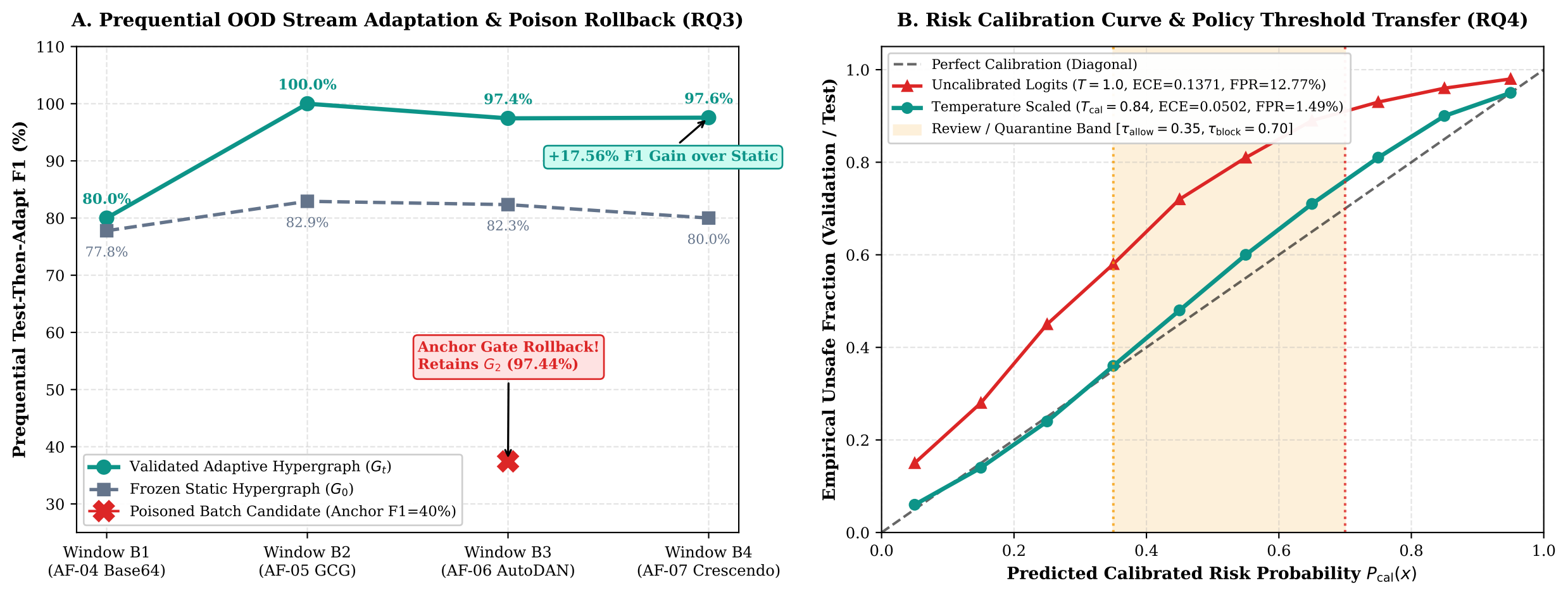

A. Prequential OOD Stream Adaptation & Poison Rollback (RQ3)
Prequential Test-Then-Adapt F1 (%)
110
100
90
80
70
60
50
40
30
100.0%
97.4%
97.6%
80.0%
82.9%
82.3%
80.0%
77.8%
+17.56% F1 Gain over Static
Anchor Gate Rollback!
Retains G_2 (97.44%)
Validated Adaptive Hypergraph (G_t)
Frozen Static Hypergraph (G_0)
Poisoned Batch Candidate (Anchor F1=40%)
Window B1
(AF-04 Base64)
Window B2
(AF-05 GCG)
Window B3
(AF-06 AutoDAN)
Window B4
(AF-07 Crescendo)
B. Risk Calibration Curve & Policy Threshold Transfer (RQ4)
Empirical Unsafe Fraction (Validation / Test)
1.0
0.8
0.6
0.4
0.2
0.0
Perfect Calibration (Diagonal)
Uncalibrated Logits ($T = 1.0$, ECE=0.1371, FPR=12.77%)
Temperature Scaled ($T_{cal} = 0.84$, ECE=0.0502, FPR=1.49%)
Review / Quarantine Band [$\tau_{allow} = 0.35$, $\tau_{block} = 0.70$]
0.0
0.2
0.4
0.6
0.8
1.0
Predicted Calibrated Risk Probability $P_{cal}(x)$